Probability and Stochastic Processes

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Chapter 6 Discrete Time Markov Chains

Many systems represent the simplest generalization of independent processes and have the property that the future is independent of the past given the present. This property is called the ***Markov property***, and systems having this property are called ***Markov processes***. If the index set is discrete, we call it ***Markov chain***. They widely appear both in pure and applied mathematics, and have many applications in science and technology.

In this chapter we give a short survey of the properties of Markov chains.

Contents

1. Basic Concepts
2. Markov Chains Having Two States
3. Higher Order Transition Probabilities and Distributions
4. Invariant Distributions and Ergodic Markov Chain

Basic Concepts

Example

A frog hops about on 7 lily pads. The numbers next to arrows show the probabilities with which, at the next jump, he jumps to a neighboring lily pad (and when out-going probabilities sum to less than 1 he stays where he is with the remaining probability). There are 7 "states" (lily pads).
We would like to know where do we go, how long does it take to get there, and what happens in the long run? Specifically:

Basic Concepts

Example

(a) Starting in state 1, what is the probability that we are still in state 1 after 3 steps? (1/4) after 5 steps? (3/16) or after 1000 steps? ($\approx 1/5$)
(b) Starting in state 4, what is the probability that we ever reach state 7? (1/3)
(c) Starting in state 4, how long on average does it take to reach either 3 or 7? (11/3)
(d) Starting in state 2, what is the long-run proportion of time spent in state 3? (2/5)

Basic Concepts

Markov chains models/methods are useful in answering questions such as:

How long does it take to shuffle deck of cards?

How likely is a queue to overflow its buffer?

How long does it take for a knight making random moves on a chessboard to return to his initial square?

What do the hyperlinks between web pages say about their relative popularity?

Basic Concepts

Definition

Suppose that $\{X_n\}_{n\geqslant 0}$ *is a stochastic process with the countable state space* S*. For all* $n \geqslant 0$*,* $P\{X_n \in S\} = 1$*.* $\{X_n\}_{n\geqslant 0}$ *is said to be a* ***discrete-time Markov chain****, if it satisfies the Markov property:*

$$\begin{aligned} P(X_{n+1} = y | X_n = x, X_{n-1} = x_{n-1}, \ldots, X_0 = x_0) \\ = P(X_{n+1} = y | X_n = x). \end{aligned}$$

for all $n \geq 0$ *and all* $x_0, x_1, \ldots, x_{n-1}, x, y \in S$*.*

Generally, we can assume the state space of a Markov chain is $S = \{0, 1, \cdots, N\}$ if the original state space is finite, and $S = \{0, 1, 2 \cdots\}$ if the original state space is countable infinite.

Basic Concepts

The conditional probabilities

$$P(X_{n+1} = y | X_n = x), \quad \forall n \geqslant 0, \ \forall x, y \in S$$

are called the ***one-step transition probabilities*** of the Markov chain.

Furthermore, the Markov chain is said to be ***time homogeneous***, or to have stationary transition probabilities if for any $x, y \in S$,

$$P(X_{m+1} = y | X_m = x) = P(X_{n+1} = y | X_n = x) \tag{1}$$

holds for any $m, n \geqslant 0$.

Basic Concepts

Let

$$p_{xy}(n, n+1) := P(X_{n+1} = y | X_n = x).$$

Then

$$p_{xy}(0, 1) = p_{xy}(n, n+1)$$

if the Markov chain is time homogeneous. We call $p_{xy}(0, 1)$ the ***one-step transition probability***, which represents the probability that the process will, when in state x, next make a transition into state y. We usually use the abbreviations $p_{xy}(1)$ or even p_{xy} to replace $p_{xy}(0, 1)$.

It is obvious that

$$p_{xy} \geqslant 0, \ x, y \geqslant 0; \qquad \sum_{y \in S} p_{xy} = 1, \quad \forall x \in S.$$

Basic Concepts

To describe all transition probabilities more properly and orderly, let $\mathbf{P} := \mathbf{P}(1) = (p_{xy})$ denote the matrix of one-step transition probabilities, that is

$$\mathbf{P} = \mathbf{P}(1) = (p_{xy}) = \begin{pmatrix} p_{00} & p_{01} & \cdots & p_{0N} \\ p_{10} & p_{11} & \cdots & p_{1N} \\ \vdots & \vdots & \ddots & \vdots \\ p_{N0} & p_{N1} & \cdots & p_{NN} \end{pmatrix} \text{ if } S = \{0, 1, \cdots, N\},$$

Basic Concepts

$$\mathbf{P} = \mathbf{P}(1) = (p_{ij}) = \begin{pmatrix} p_{00} & p_{01} & \cdots & p_{0j} & \cdots \\ p_{10} & p_{11} & \cdots & p_{1j} & \cdots \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ p_{i0} & p_{i1} & \cdots & p_{ij} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \text{ if } S = \{0, 1, 2, \cdots\},$$

and $\mathbf{P}$ is called ***transition*** (***probability***) ***matrix*** of one-step transition probabilities. Observe that row sums of transition matrix equal to one. In fact, any nonnegative matrix with row sums equal to one is called a ***stochastic matrix***. (Notice that the state space is sometimes written on the left of the matrix to help in keeping track of which row refers to which state.)

Basic Concepts

Example

Consider a farmer using an old tractor. The tractor is often in the repair shop but it always takes only one day to get it running again. The first day out of the shop it always works but on any given day thereafter, independent of its previous history, there is a 10% chance of it not working and thus being sent back to the shop. Let $X_0, X_1, \cdots$ be random variables denoting the daily condition of the tractor, where a one denotes the working condition and a zero denotes the failed condition. In other words, $X_n = 1$ denotes that the tractor is working on day n and $X_n = 0$ denotes it being in the repair shop on day n. Thus, $X_0, X_1, \cdots$ is a sequence of random variables. $\square$

Basic Concepts

Example

(**A communications system**) Consider a communications system that transmits the digits 0 and 1. Each digit transmitted must pass through several stages, at each of which there is a probability p that the digit entered will be unchanged when it leaves. Let X_n denote the digit entering the nth stage. Show that $\{X_n\}$ is a Markov chain. Is this Markov chain homogeneous? Determine the one-step transition probability matrix.

Basic Concepts

Example

A salesman lives in town A and is responsible for towns A, B and C. Each week he is required to visit a different town. When he is in his home town, it makes no difference which town he visits next so he flips a coin and if it is heads he goes to B and if tails he goes to C. However, after spending a week away from home he has a slight preference for going home so when he is in either towns B or C he flips two coins. If two heads occur, then he goes to the other town; otherwise he goes to A. Let the random variable X_n equals A, B, or C according to his location during week n. Show that $\{X_n\}_{n\geqslant 0}$ is a Markov chain. Is this Markov chain homogeneous? Determine the one-step transition probability matrix.

Basic Concepts

Example

Suppose that $\{X_n\}_{n\geqslant 0}$ is an IID random sequence, each of which may take the values from $S = \{0, 1, 2, \cdots, N\}$. Show that $\{X_n\}$ is a Markov chain. Is this Markov chain homogeneous? Determine the one-step transition probability.

Definition

(***A random walk model***) *A Markov chain whose state space is given by the integers* $x = 0, \pm 1, \pm 2, \cdots$ *is said to be a random walk if, for some* $0 < p < 1$,

$$p_{xx+1} = p = 1 - p_{xx-1}, \qquad x = 0, \pm 1, \cdots .$$

Basic Concepts

Example

(**A gambling model**) Consider a gambler who, at each play of the game, either wins 1 with probability p or loses 1 with probability $q = 1 - p$. If we suppose that our gambler quits playing either when he goes broke or he attains a fortune of N, then the gambler's fortune is a Markov chain having a transition probabilities

$$p_{x\ x+1} = p = 1 - p_{x\ x-1}, \qquad x = 1, 2, \cdots, N-1,$$

$$P_{00} = P_{NN} = 1.$$

States 0 and N are called *absorbing states* since once entered they are never left. Notice that the preceding is a finite state random walk with absorbing barriers (states 0 and N).

Basic Concepts

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 & 0 & \cdots & \cdots & 0 & 0 & 0 & 0 \\ q & 0 & p & 0 & \cdots & \cdots & 0 & 0 & 0 & 0 \\ 0 & q & 0 & p & \cdots & \cdots & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \ddots & \ddots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \cdots & q & 0 & p & 0 \\ 0 & 0 & 0 & 0 & \cdots & \cdots & 0 & q & 0 & p \\ 0 & 0 & 0 & 0 & \cdots & \cdots & 0 & 0 & 0 & 1 \end{pmatrix}.$$

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Markov Chains Having Two States

For a Markov chain, we concern the following three basic questions.

(i) For a given time n, what is the distribution of X_n?

(ii) If n is large enough, what is the limit distribution of X_n?

(iii) For k given times $n_1 < n_2 < \cdots < n_k$, what is the joint distribution of the random vector $(X_{n_1}, X_{n_2}, \cdots, X_{n_k})$?

We will answer the above three questions for the example of weather forecasting. This is an example of a Markov chain only having two states.

Markov Chains Having Two States

Suppose that the chance of rain tomorrow depends on previous weather conditions only through whether or not it is raining today and not on past weather conditions. Suppose also that if it rains today, then it will not rain tomorrow with probability α; and if it does not rain today, then it will rain tomorrow with probability β.

Markov Chains Having Two States

Let X_n be random variables denoting the weather condition of day n. If we say that the process is in state 0 when it rains and state 1 when it does not rain, then $X_0, X_1, \cdots$ is a random sequence with state space $S = \{0, 1\}$. For all n,

$$P(X_{n+1} = 1 | X_n = 0, X_{n-1}$$

$$= x_{n-1}, \ldots, X_0 = x_0) = P(X_{n+1} = 1 | X_n = 0) = \alpha,$$

where $x_i = 0$ or 1, $i = 0, 1, \cdots, n-1$. Similarly,

$$\begin{aligned} P(X_{n+1} = 0 \,|X_n = 0) &= 1 - \alpha, \\ P(X_{n+1} = 0 \,|X_n = 1) &= \beta, \\ P(X_{n+1} = 1 \,|X_n = 1) &= 1 - \beta. \end{aligned}$$

Thus, $\{X_n\}_{n \geqslant 0}$ is a Markov chain with state space $S = \{0, 1\}$.

Markov Chains Having Two States

In fact, it is also homogeneous and has the transition matrix

$$\mathbf{P} = \begin{pmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{pmatrix} = \begin{pmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{pmatrix}.$$

To answer the above questions, we need a necessary information about the initial probability. Let $p_0(0) = P(X_0 = 0)$. Then

$$p_1(0) = P(X_0 = 1) = 1 - p_0(0).$$

Markov Chains Having Two States

From this information, we can easily compute $P(X_n = 0)$ and $P(X_n = 1)$. We observe that

$$\begin{aligned}
P(X_{n+1} = 0) &= P(X_{n+1} = 0, X_n = 0) + P(X_{n+1} = 0, X_n = 1) \\
&= P(X_{n+1} = 0 | X_n = 0) P(X_n = 0) \\
&\quad + P(X_{n+1} = 0 | X_n = 1) P(X_n = 1) \\
&= (1 - \alpha) P(X_n = 0) + \beta P(X_n = 1) \\
&= (1 - \alpha) P(X_n = 0) + \beta (1 - P(X_n = 0)) \\
&= (1 - \alpha - \beta) P(X_n = 0) + \beta.
\end{aligned}$$

Markov Chains Having Two States

Now $P(X_0 = 0) = p_0(0)$, thus

$$P(X_1 = 0) = (1 - \alpha - \beta)p_0(0) + \beta$$

and

$$\begin{aligned} P(X_2 = 0) &= (1 - \alpha - \beta)P(X_1 = 0) + \beta \\ &= (1 - \alpha - \beta)^2 p_0(0) + \beta[1 + (1 - \alpha - \beta)]. \end{aligned}$$

It is easily seen by repeating this procedure n times that

$$P(X_n = 0) = (1 - \alpha - \beta)^n p_0(0) + \beta \sum_{i=0}^{n-1} (1 - \alpha - \beta)^i. \qquad (2)$$

In the trivial case $\alpha = \beta = 0$, it is clear that for all n

$$P(X_n = 0) = p_0(0) \quad \text{and} \quad P(X_n = 1) = p_1(0).$$

Markov Chains Having Two States

Suppose that $\alpha + \beta > 0$. Then by the formula for the sum of a finite geometric progression,

$$\sum_{i=0}^{n-1} (1 - \alpha - \beta)^i = \frac{1 - (1 - \alpha - \beta)^n}{\alpha + \beta}.$$

We conclude from equation (2) that

$$P(X_n = 0) = \frac{\beta}{\alpha + \beta} + (1 - \alpha - \beta)^n \left(p_0(0) - \frac{\beta}{\alpha + \beta} \right), \qquad (3)$$

and consequently that

$$P(X_n = 1) = \frac{\alpha}{\alpha + \beta} + (1 - \alpha - \beta)^n \left(p_1(0) - \frac{\beta}{\alpha + \beta} \right). \qquad (4)$$

Markov Chains Having Two States

Suppose that α and β are neither equal to 0 nor equal to 1. Then we have $0 < \alpha + \beta < 2$, which applies that $|1 - \alpha - \beta| < 1$. In this case,

$$(1 - \alpha - \beta)^n \to 0 \text{ as } n \to \infty.$$

We can conclude that

$$\lim_{n\to\infty} P(X_n = 0) = \frac{\beta}{\alpha + \beta} \quad \text{and} \quad \lim_{n\to\infty} P(X_n = 1) = \frac{\alpha}{\alpha + \beta}. \tag{5}$$

Markov Chains Having Two States

Let us choose another expression to describe the distribution of X_n. Denote

$$p_x(n) := P(X_n = x),$$

where $x = 0$ and 1. Let $p(n) = \big(p_0(n), p_1(n)\big)$. We can see if $n = 0$, then $p(0)$ is the initial distribution. If $n = 1$, then

$$\begin{aligned} P(X_1 = 0) &= P(X_0 = 0)P(X_1 = 0|X_0 = 0) \\ &\quad + P(X_0 = 1)P(X_1 = 0|X_0 = 1) \\ &= p_0(0)p_{00} + p_1(0)p_{10} = \big(p_0(0), p_1(0)\big) \begin{pmatrix} p_{00} \\ p_{10} \end{pmatrix}. \end{aligned}$$

Similarly,

$$P(X_1 = 1) = \big(p_0(0), p_1(0)\big) \begin{pmatrix} p_{01} \\ p_{11} \end{pmatrix}.$$

Then we have

$$p(1) = (p_0(1), p_1(1)) = p(0) \begin{pmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{pmatrix} = p(0)\mathbf{P}.$$

Markov Chains Having Two States

By a diagonalization argument we know

$$\mathbf{P}^n = \frac{1}{\alpha+\beta}\begin{pmatrix} \beta+\alpha(1-\alpha-\beta)^n & \alpha(1-(1-\alpha-\beta)^n) \\ \beta(1-(1-\alpha-\beta)^n) & \alpha+\beta(1-\alpha-\beta)^n \end{pmatrix}, \tag{6}$$

Then

$$\begin{aligned} & p(n) = p(0)\mathbf{P}^n \\ = & (p_0(0), p_1(0))\cdot \\ & \frac{1}{\alpha+\beta}\begin{pmatrix} \beta+\alpha(1-\alpha-\beta)^n & \alpha(1-(1-\alpha-\beta)^n) \\ \beta(1-(1-\alpha-\beta)^n) & \alpha+\beta(1-\alpha-\beta)^n \end{pmatrix} \\ = & (\frac{\beta}{\alpha+\beta} + (1-\alpha-\beta)^n\left(p_0(0) - \frac{\beta}{\alpha+\beta}\right), \\ & \frac{\alpha}{\alpha+\beta} + (1-\alpha-\beta)^n\left(p_1(0) - \frac{\beta}{\alpha+\beta}\right)), \end{aligned}$$

Markov Chains Having Two States

which is coincidence with equations (3) and (4).

But what is the meaning of $\mathbf{P}^n$?

Now we discuss how to answer the third question. If we know the initial distribution and one-step transition matrix, then it is enough to show how to compute the quantities

$$P(X_0 = x_0, X_1 = x_1, \cdots X_n = x_n),$$

where $x_i = 0$ or 1, $i = 0, 1, 2, \cdots, n$.

Markov Chains Having Two States

By the definition of conditional probabilities we obtain

$$\begin{aligned} &P(X_0 = x_0, X_1 = x_1, \cdots, X_n = x_n) \\ &= P(X_n = x_n | X_0 = x_0, X_1 = x_1, \cdots, X_{n-1} = x_{n-1}) \\ &\qquad \cdot P(X_0 = x_0, X_1 = x_1, \cdots, X_{n-1} = x_{n-1}) \end{aligned}$$

Then

$$\begin{aligned} &P(X_n = x_n | X_0 = x_0, X_1 = x_1, \cdots, X_{n-1} = x_{n-1}) \\ &= P(X_n = x_n | X_{n-1} = x_{n-1}) = p_{x_{n-1} x_n} \end{aligned}$$

by using Markov property. Thus

$$\begin{aligned} &P(X_0 = x_0, X_1 = x_1, \cdots, X_n = x_n) \\ &= p_{x_{n-1} x_n} \cdot P(X_0 = x_0, X_1 = x_1, \cdots, X_{n-1} = x_{n-1}). \end{aligned}$$

Markov Chains Having Two States

If we proceed by induction, then

$$\begin{aligned} &P(X_0 = x_0, X_1 = x_1, \cdots, X_n = x_n) \\ &= p_{x_{n-1}x_n} p_{x_{n-2}x_{n-1}} \cdots p_{x_0 x_1} p_{x_0}(0). \end{aligned}$$

For example, we can easily obtain that

$$P(X_0 = 0, X_1 = 1 \text{ and } X_2 = 0) = p_0(0) p_{01} p_{10} = p_0(0)\alpha\beta.$$

But for the probability $P(X_0 = 1, X_2 = 1 \text{ and } X_3 = 0)$, it is not a simple work.

Markov Chains Having Two States

$$
\begin{aligned}
& P(X_0 = 1, X_2 = 1 \text{ and } X_3 = 0) \\
= & P(X_0 = 1)P(X_2 = 1|X_0 = 1)P(X_3 = 0|X_2 = 1, X_1 = 1) \\
= & P(X_0 = 1)P(X_2 = 1|X_0 = 1)P(X_3 = 0|X_2 = 1) \\
= & p_1(0)P(X_2 = 1|X_0 = 1)p_{10}.
\end{aligned}
$$

We can not obtain the probability $P(X_2 = 1|X_0 = 1)$ directly from the one-step transition probability matrix. We need use the following total law of probability of conditional probability. Suppose that A, B and C are three events, and $P(BC) > 0$ $P(\overline{B}C) > 0$.

Markov Chains Having Two States

Then

$$\begin{aligned} P(A|C) &= P(AB \cup A\overline{B}|C) \\ &= P(AB|C) + P(A\overline{B}|C) \\ &= P(B|C)P(A|BC) + P(\overline{B}|C)P(A|\overline{B}C) \end{aligned}$$

By letting $A = \{X_2 = 1\}$, $B = \{X_1 = 0\}$ and $C = \{X_0 = 1\}$, we have

$$\begin{aligned} P(X_2 = 1|X_0 = 1) &= P(X_2 = 1, X_1 = 0|X_0 = 1) \\ &\quad + P(X_2 = 1, X_1 = 1|X_0 = 1) \\ &= P(X_1 = 0|X_0 = 1)P(X_2 = 1|X_1 = 0, X_0 = 1) \\ &\quad + P(X_1 = 1|X_0 = 1)P(X_2 = 1|X_1 = 1, X_0 = 1) \\ &= P(X_1 = 0|X_0 = 1)P(X_2 = 1|X_1 = 0) \\ &\quad + P(X_1 = 1|X_0 = 1)P(X_2 = 1|X_1 = 1) \\ &= p_{10}p_{01} + p_{11}p_{11}. \end{aligned}$$

Markov Chains Having Two States

Thus

$$\begin{aligned} P(X_0 = 1, X_2 = 1 \text{ and } X_3 = 0) &= p_1(0)(p_{10}p_{01} + p_{11}p_{11})p_{10} \\ &= p_1(0)p_{10}p_{01}p_{10} + p_0(1)p_{11}p_{11}p_{10}. \end{aligned}$$

That means all circumstance of intermediate steps should be taken into account.

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Higher Order Transition Probabilities and Distributions

In this section, we find the ditributions for general discrete-time Markov chains.
Let

$$p_x(0) = P(X_0 = x), \quad x \in S.$$

be the ***initial probability distribution*** of X_0. An initial probability distribution can be given by a vector

$$p(0) = (p_x(0))_{x \in S} = (p_0(0), p_1(0), \cdots).$$

Higher Order Transition Probabilities and Distributions

The probabilities

$$P(X_{n+m} = y | X_n = x), \qquad n \geqslant 0, m \geqslant 1 \;\; x, y \in S$$

are called the m**-step transition probabilities**. We denote it by $p_{xy}(m)$. i.e.,

$$p_{xy}(m) = P(X_m = y | X_0 = x) = P(X_{n+m} = y | X_n = x).$$

The latter equality holds because of time homogeneity. Let $\mathbf{P}(m) = (p_{xy}(m))$ denote the m**-step transition probability matrix**, that is

Higher Order Transition Probabilities and Distributions

$$\mathbf{P}(m) = (p_{xy}(m)) = \begin{pmatrix} p_{00}(m) & p_{01}(m) & \cdots & p_{0N}(m) \\ p_{10}(m) & p_{11}(m) & \cdots & p_{1N}(m) \\ \vdots & \vdots & \ddots & \vdots \\ p_{N0}(m) & p_{N1}(m) & \cdots & p_{NN}(m) \end{pmatrix}$$

for $S = \{0, 1, \cdots, N\}$.

Higher Order Transition Probabilities and Distributions

By using the information of initial probability distribution and the m-step transition probabilities, the distribution of X_n can be given by

$$\begin{aligned} P(X_n = y) &= \sum_x P(X_n = y | X_0 = x) P(X_0 = x) \\ &= \sum_x p_x(0) p_{xy}(n), \quad \forall y \in S. \end{aligned}$$

Let $p_y(n) = P(X_n = y)$ and $p(n) = (p_0(n), p_1(n), p_2(n), \cdots)$. Then the distribution of X_n asserts that

$$p(n) = p(0)\mathbf{P}(n). \tag{7}$$

That is, the one-dimensional distribution can be determined by the initial distribution and the n-step transition probability matrix.

Higher Order Transition Probabilities and Distributions

We will now show that the n-step transition probability is in fact the (x, y) entry in the matrix $\mathbf{P}^n$. To see this, we first note that this is trivially true for $n = 1$. Assume it is true for a given n.

$$
\begin{aligned}
& P(X_{n+1} = y \mid X_0 = x) \\
= & \sum_{z \in S} P(X_{n+1} = y, X_n = z \mid X_0 = x) \\
= & \sum_{z \in S} P(X_n = z | X_0 = x) \cdot P(X_{n+1} = y | X_n = z, X_0 = x) \\
= & \sum_{z \in S} P(X_n = z | X_0 = x) \cdot P(X_{n+1} = y | X_n = z) \\
= & \sum_{z \in S} p_{xz}(n) p_{zy}, \quad x, y \in S
\end{aligned}
$$

Higher Order Transition Probabilities and Distributions

But if $p_{xz}(n)$ is the (x, z) entry of P^n, then the last sum is exactly the (x, y) entry of $\mathbf{P}^n\mathbf{P} = \mathbf{P}^{n+1}$. That is for any $n \geqslant 1$,

$$\mathbf{P}(n) = \mathbf{P}^n.$$

Thus n-step transition probabilities can be determined by one-step transition probabilities. Then equations (7) changes to

$$p(n) = p(0)\mathbf{P}(n) = p(0)\mathbf{P}^n, \tag{8}$$

which means the one-dimensional distribution of a Markov chain can be determined by the initial distribution and the one-step transition probability matrix.

Higher Order Transition Probabilities and Distributions

Next let us discuss the n-dimensional distribution of a Markov chain. For any positive integer k, for all $0 \leqslant n_1 < n_2 < \cdots < n_k$ and for all $x_1, x_2, \cdots, x_k \in S$,

$$
\begin{aligned}
& P(X_{n_1} = x_1, X_{n_2} = x_2, \cdots, X_{n_k} = x_k) \\
= & P(X_{n_1} = x_1) \cdot P(X_{n_2} = x_2 | X_{n_1} = x_1) \cdot \\
& \qquad \cdots \cdot P(X_{n_k} = x_k | X_{n_1} = x_1, X_{n_2} = x_2, \ldots, X_{n_{k-1}} = x_{k-1}) \\
= & P(X_{n_1} = x_1) P(X_{n_2} = x_2 | X_{n_1} = x_1) \cdots P(X_{n_k} = x_k | X_{n_{k-1}} = x_{k-1}) \\
= & p_{x_1}(n_1) p_{x_1 x_2}(n_2 - n_1) \cdots p_{x_{k-1} x_k}(n_k - n_{k-1}).
\end{aligned}
$$

Since $p(n)$ and $p_{xy}(m)$ are all determined by $p(0)$ and the one-step transition matrix $\mathbf{P}$, we have the following theorem:

Higher Order Transition Probabilities and Distributions

Theorem

The discrete-time Markov chain $\{X_n, n = 0, 1, 2, \ldots\}$ *can be uniquely determined by its initial distribution* $p(0)$ *and the one-step transition probability matrix* $\mathbf{P}$.

Theorem

Suppose that $\{X_n\}$ *is a homogeneous Markov chain. Then*

$$p_{xy}(n+m) = \sum_z p_{xz}(n) p_{zy}(m), \quad for\ all\ m, n \geq 0, all\ x, y.$$

These equations are called ***Chapman-Kolmogorov (C-K) equations***.

Higher Order Transition Probabilities and Distributions

Example

Consider the weather forecasting example in section 6.2 in which the weather is considered as a two-state Markov chain. Let $\alpha = 0.3$ and $\beta = 0.4$. Calculate the probability that it will rain four days from today given that it is raining today.

Higher Order Transition Probabilities and Distributions

Solution. The state space is $S = \{0, 1\}$. What we want is the probability $p_{00}(4)$. There are 8 transition ways:

$$0 \xrightarrow{1-\alpha} 0 \xrightarrow{1-\alpha} 0 \xrightarrow{1-\alpha} 0 \xrightarrow{1-\alpha} 0; \qquad 0 \xrightarrow{1-\alpha} 0 \xrightarrow{1-\alpha} 0 \xrightarrow{\alpha} 1 \xrightarrow{\beta} 0;$$

$$0 \xrightarrow{1-\alpha} 0 \xrightarrow{\alpha} 1 \xrightarrow{\beta} 0 \xrightarrow{1-\alpha} 0; \qquad 0 \xrightarrow{1-\alpha} 0 \xrightarrow{\alpha} 1 \xrightarrow{1-\beta} 1 \xrightarrow{\beta} 0;$$

$$0 \xrightarrow{\alpha} 1 \xrightarrow{\beta} 0 \xrightarrow{1-\alpha} 0 \xrightarrow{1-\alpha} 0; \qquad 0 \xrightarrow{\alpha} 1 \xrightarrow{1-\beta} 1 \xrightarrow{\beta} 0 \xrightarrow{1-\alpha} 0;$$

$$0 \xrightarrow{\alpha} 1 \xrightarrow{\beta} 0 \xrightarrow{\alpha} 1 \xrightarrow{\beta} 0; \qquad 0 \xrightarrow{\alpha} 1 \xrightarrow{1-\beta} 1 \xrightarrow{1-\beta} 1 \xrightarrow{\beta} 0.$$

Thus

$$\begin{aligned} p_{00}(4) &= (1-\alpha)^4 + 3(1-\alpha)^2\alpha\beta + 2\alpha\beta(1-\alpha)(1-\beta) + \\ &\quad \alpha^2\beta^2 + \alpha\beta(1-\beta)^2 \\ &\overset{\alpha=0.3,\beta=0.4}{=} 0.2401 + 0.1764 + 0.1008 + 0.0144 + 0.0432 = 0.5749. \end{aligned}$$

Higher Order Transition Probabilities and Distributions

Alternatively, since the one-step transition matrix is given by $\mathbf{P} = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix}$,

$$\mathbf{P}(2) = \mathbf{P}^2 = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix} \cdot \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix} = \begin{pmatrix} 0.61 & 0.39 \\ 0.52 & 0.48 \end{pmatrix},$$

$$\mathbf{P}(4) = (\mathbf{P}^2)^2 = \begin{pmatrix} 0.61 & 0.39 \\ 0.52 & 0.48 \end{pmatrix}^2) = \begin{pmatrix} 0.5749 & 0.4251 \\ 0.5668 & 0.4332 \end{pmatrix}.$$

The desired probability $p_{00}(4)$ equals 0.5749. $\square$

Higher Order Transition Probabilities and Distributions

Example

(**A model of class mobility**) A problem of interest to sociologists is to determine the proportion of society that has an upper- or lower-class occupation. One possible mathematical model would be to assume that transitions between social classes of the successive generations in a family can be regarded as transitions of a Markov chain. Assume that the occupation of a child depends only on his or her parent's occupation. Let us suppose that such a Markov model is appropriate and that the transition probability matrix is given by

$$\mathbf{P} = \begin{pmatrix} 0.45 & 0.48 & 0.07 \\ 0.05 & 0.70 & 0.25 \\ 0.01 & 0.50 & 0.49 \end{pmatrix}.$$

Higher Order Transition Probabilities and Distributions

Let 0 be upper-class occupation, 1 middle-class occupation and 2 lower-class occupation. The state space is $S = \{0, 1, 2\}$. Suppose a parent is in state 2(the lower class), what is the probability that a grand child will be in state 1(the middle class)?

Higher Order Transition Probabilities and Distributions

The one-step transition matrix of a Markov chain always plays a much greater role in describing its properties than does the initial distribution. For this reason, it is customary to study simultaneously all Markov chains having a given transition matrix. In fact, we adhere to the usual convention that by "a Markov chain having a transition matrix $\mathbf{P}$", we really mean the family of all Markov chains having that transition matrix.

Higher Order Transition Probabilities and Distributions

Example

Consider the homogeneous Markov chain $\{X_n, n \geq 0\}$ consisting of three states $0, 1, 2$ and having transition probability matrix

$$\begin{pmatrix} \frac{3}{4} & \frac{1}{4} & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{3}{4} & \frac{1}{4} \end{pmatrix}.$$

The initial distribution is given by $p_i(0) = \frac{1}{3}$, $i = 0, 1, 2$. Find the probabilities $P(X_0 = 0, X_2 = 1)$ and $P(X_2 = 1)$.

Higher Order Transition Probabilities and Distributions

Solution. It is easy to calculate that the 2-step transition probability matrix is

$$\mathbf{P}(2) = \mathbf{P}^2 = \begin{pmatrix} \frac{3}{4} & \frac{1}{4} & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{3}{4} & \frac{1}{4} \end{pmatrix} \begin{pmatrix} \frac{3}{4} & \frac{1}{4} & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{3}{4} & \frac{1}{4} \end{pmatrix} = \begin{pmatrix} \frac{5}{8} & \frac{5}{16} & \frac{1}{16} \\ \frac{5}{16} & \frac{1}{2} & \frac{3}{16} \\ \frac{3}{16} & \frac{9}{16} & \frac{1}{4} \end{pmatrix}.$$

Thus,

$$P(X_0 = 0, X_2 = 1) = p_0(0)p_{01}(2) = \frac{1}{3} \times \frac{5}{16} = \frac{5}{48},$$

and

$$P(X_2 = 1) = p_0(0)p_{01}(2) + p_1(0)p_{11}(2) + p_2(0)p_{21}(2) = \frac{11}{24}.$$

$\square$

Higher Order Transition Probabilities and Distributions

Example

An urn always contains 2 balls. Ball colors are red and blue. At each stage a ball is randomly chosen and then replaced by a new ball, which with probability 0.8 is the same color, and with probability 0.2 is the opposite color, as the ball it replaces. Suppose initially both balls are red. Find the probability that the fifth ball selected is red.

Contents

1. Basic Concepts
2. Markov Chains Having Two States
3. Higher Order Transition Probabilities and Distributions
4. Invariant Distributions and Ergodic Markov Chain

Invariant Distributions and Ergodic Markov Chain

Question

What is the limiting distribution of X_n as $n \to \infty$?

Since $p(n) = p(0)\mathbf{P}^n$, we consider the property of $\mathbf{P}^n$ as $n \to \infty$.

For the example of weather forecasting in section 6.2, we have

$$\lim_{n\to\infty} \mathbf{P}^n = \lim_{n\to\infty} \begin{pmatrix} p_{00}(n) & p_{01}(n) \\ p_{10}(n) & p_{11}(n) \end{pmatrix} = \begin{pmatrix} \frac{\beta}{\alpha+\beta} & \frac{\alpha}{\alpha+\beta} \\ \frac{\beta}{\alpha+\beta} & \frac{\alpha}{\alpha+\beta} \end{pmatrix}.$$

Let

$$\pi = (\pi_0, \pi_1) = \left(\frac{\beta}{\alpha+\beta}, \frac{\alpha}{\alpha+\beta}\right). \tag{9}$$

That means for n large enough,

$$\mathbf{P}^n \approx \begin{pmatrix} \pi \\ \pi \end{pmatrix}.$$

Invariant Distributions and Ergodic Markov Chain

Let

$$\pi = (\pi_0, \pi_1) = \left(\frac{\beta}{\alpha + \beta}, \frac{\alpha}{\alpha + \beta} \right). \tag{10}$$

That means for n large enough,

$$\mathbf{P}^n \approx \begin{pmatrix} \pi \\ \pi \end{pmatrix}.$$

We call π a limiting distribution as n goes to infinity, provided $(\alpha, \beta) \neq (1, 1)$. Whatever the initial state X_0, the probability of being at state 1 after a "large" time becomes close to $\alpha/(\alpha + \beta)$, while the probability of being at state 0 becomes close to $\beta/(\alpha + \beta)$.

Invariant Distributions and Ergodic Markov Chain

We also note that the distribution π in (10) is invariant by $\mathbf{P}$ in the sense that

$$\begin{aligned}\pi\mathbf{P} &= \left(\frac{\beta}{\alpha+\beta}, \frac{\alpha}{\alpha+\beta}\right) \cdot \begin{pmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{pmatrix} \\ &= \frac{1}{\alpha+\beta}\Big(\beta(1-\alpha)+\alpha\beta,\ \alpha\beta+\alpha(1-\beta)\Big) = \pi\end{aligned}$$

i.e., π satisfies the invariance relation $\pi = \pi\mathbf{P}$. We call a probability vector π an ***invariant*** (***probability***) ***distribution*** for $\mathbf{P}$ if

$$\pi = \pi\mathbf{P}.$$

Invariant Distributions and Ergodic Markov Chain

If we let π be the initial distribution, i.e., $p(0) = \pi$, then

$$p(n) = p(0)\mathbf{P}^n = \pi\mathbf{P}^n = \pi.$$

Such a π is also called a *stationary*, *equilibrium*, or *steady-state probability distribution*.

There are two natural questions to ask about invariant probability distributions for stochastic matrices:

(1) Does every stochastic matrix $\mathbf{P}$ have an invariant probability distribution π?

(2) Is the invariant probability distribution unique?

Invariant Distributions and Ergodic Markov Chain

Notice that if there exists π such that

$$\lim_{n\to\infty} \mathbf{P}^n = \begin{pmatrix} \pi \\ \pi \\ \vdots \\ \pi \end{pmatrix},$$

then for all initial probability distributions $p(0)$,

$$\lim_{n\to\infty} p(0)\mathbf{P}^n = \pi.$$

Invariant Distributions and Ergodic Markov Chain

Definition

Consider a Markov chain $\{X_n\}$ *consisting of the state space* S *and having the transition probability matrix* $\mathbf{P}(n) = (p_{xy}(n))$. *If for all* $x, y \in S$, $\lim_{n\to\infty} p_{xy}(n)$ *exists and is independent of* x, *that is,*

$$\lim_{n\to\infty} p_{xy}(n) = \pi_y,$$

or

$$\lim_{n\to\infty} \mathbf{P}(n) = \begin{pmatrix} \pi_1 & \pi_2 & \cdots & \pi_j & \cdots \\ \pi_1 & \pi_2 & \cdots & \pi_j & \cdots \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \pi_1 & \pi_2 & \cdots & \pi_j & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix},$$

then we say that the Markov chain is ***ergodic****. Furthermore, if* $\sum_x \pi_x = 1$, *then* $\pi = (\pi_1, \pi_2 \cdots)$ *is called the limiting probability distribution.*

Invariant Distributions and Ergodic Markov Chain

Theorem

Consider a homogeneous Markov chain $\{X_n\}$ *consisting of the state space* $S = \{0, 1, 2, \cdots, N\}$ *and having transition probability matrix* $\mathbf{P}$. *If*

$$\text{there exists } m > 0, \text{ such that for all } x, y \in S, \ p_{xy}(m) > 0, \tag{11}$$

then the Markov chain is ergodic and it has the limiting distribution $\pi = (\pi_0, \pi_1, \pi_2 \ldots, \pi_N)$ *(i.e., invariant distribution),* π_x *is the unique nonnegative solution of*

$$\begin{cases} \pi = \pi \boldsymbol{P} \\ \sum_x \pi_x = 1 \end{cases} \tag{12}$$

Invariant Distributions and Ergodic Markov Chain

Equation (12) is equivalent to

$$\begin{cases} \pi_y = \sum_{x=0}^{N} \pi_x p_{xy}, \quad where\ y = 0, 1, 2, \cdots, N, \\ \pi_0 + \pi_1 + \cdots + \pi_N = 1. \end{cases} \tag{13}$$

Invariant Distributions and Ergodic Markov Chain

Example

Consider the example in section 6.2 in which we assume that if it rains today, then it will not rain tomorrow with probability α; and if it does not rain today, then it will rain tomorrow with probability β. Find the invariant distribution.

Solution. By Theorem 3, the Markov chain has the unique invariant distribution. Then by equation (13), we have

$$\begin{aligned}\pi_0 =& \alpha\pi_0 + \beta\pi_1,\\ \pi_1 =& (1-\alpha)\pi_0 + (1-\beta)\pi_1,\\ \pi_0 +& \pi_1 = 1,\end{aligned}$$

which yields that

$$\pi_0 = \frac{\beta}{1+\beta-\alpha}, \quad \pi_1 = \frac{1-\alpha}{1+\beta-\alpha}.$$

□

Invariant Distributions and Ergodic Markov Chain

Example

Consider the model of class mobility in Example 9 again. Observe in the long run, how many people will be in upper-class?

Solution. It is obvious that X_n is ergodic. The limiting probabilities π_x thus satisfy

$$\begin{aligned}
\pi_0 &= 0.45\pi_0 + 0.05\pi_1 + 0.01\pi_2, \\
\pi_1 &= 0.48\pi_0 + 0.70\pi_1 + 0.50\pi_2, \\
\pi_2 &= 0.07\pi_0 + 0.25\pi_1 + 0.49\pi_2 \\
\pi_0 &+ \pi_1 + \pi_2 = 1
\end{aligned}$$

Hence,

$$\pi_0 = 0.07, \ \pi_1 = 0.62, \ \pi_2 = 0.31.$$

Invariant Distributions and Ergodic Markov Chain

Let $p(0)$ be the initial distribution. Then the limiting distribution of X_n is

$$\lim_{n\to\infty} p(n) = \lim_{n\to\infty} p(0)\mathbf{P}^n$$

$$= p(0)\begin{pmatrix} 0.07 & 0.62 & 0.31 \\ 0.07 & 0.62 & 0.31 \\ 0.07 & 0.62 & 0.31 \end{pmatrix} = (0.07, 0.62, 0.31).$$

That is, in the long run, 7 percent of its people in upper-class, 62 percent of its people in middle-class, and 31 percent in lower-class. $\square$

Invariant Distributions and Ergodic Markov Chain

Example

Consider a Markov chain with state space $S = \{0, 1, 2\}$ having transition probability matrix

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

Is this Markov chain ergodic?

Invariant Distributions and Ergodic Markov Chain

Solution. Since

$$\mathbf{P}(n) = \mathbf{P}^n = \mathbf{P} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix},$$

the limit of $\mathbf{P}(n)$ do exist. However, we can not find a vector π such that

$$\lim_{n\to\infty} \mathbf{P}^n = \begin{pmatrix} \pi \\ \pi \\ \pi \end{pmatrix},$$

so this chain is not ergodic. $\square$

Invariant Distributions and Ergodic Markov Chain

Example

Consider a Markov chain with state space $S = \{0, 1, 2, 3\}$ having transition probability matrix

$$\mathbf{P} = \begin{pmatrix} 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{pmatrix}.$$

Is this Markov chain ergodic?

Invariant Distributions and Ergodic Markov Chain

Solution. By calculating, we have

$$\mathbf{P}(2) = \mathbf{P}^2 = \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}$$

$$\mathbf{P}(3) = \mathbf{P}^3 = \mathbf{P}.$$

Proceeding by induction, we find that

$$\mathbf{P}(n) = \begin{cases} \mathbf{P}^2 & \text{if } n \text{ is even,} \\ \mathbf{P} & \text{if } n \text{ is odd.} \end{cases}$$

Hence, the limit of $\mathbf{P}(n)$ does not exist and this chain is not ergodic. $\square$

Summary

- **Markov Property**:

$$P(X_{n+1} = y | X_n = x, X_{n-1} = x_{n-1}, \ldots, X_0 = x_0)$$

$$= P(X_{n+1} = y | X_n = x),$$

for all $n \geq 0$ and all $x_0, x_1, \ldots, x_{n-1}, x, y \in S = \{0, 1, \cdots, N\}$.

- **Transition Probability**:

$$p_{xy} = P(X_{n+1} = y | X_n = x),$$
$$p_{xy}(m) = P(X_{n+m} = y | X_n = x).$$

- **Transition Matrix**:

$$\mathbf{P} = (p_{xy}) = \begin{pmatrix} p_{00} & p_{01} & \cdots & p_{0N} \\ p_{10} & p_{11} & \cdots & p_{1N} \\ \vdots & \vdots & \ddots & \vdots \\ p_{N0} & p_{N1} & \cdots & p_{NN} \end{pmatrix}.$$

Summary

- **Distribution of a Markov Chain**:
  - Distribution of X_n: $p(n) = (p_0(n), p_1(n), \cdots p_N(n))$

    $$p(n) = p(0)\mathbf{P}^n$$

    where $p_x(n) = P(X_n = x)$.
  - Distribution of $(X_{n_1}, X_{n_2}, \cdots, X_{n_k})$: $n_1 < n_2 < \cdots < n_k$

    $$\begin{aligned} &P(X_{n_1} = x_1, X_{n_2} = x_2, \cdots, X_{n_k} = x_k) \\ &= p_{x_1}(n_1) p_{x_1 x_2}(n_2 - n_1) \cdots p_{x_{k-1} x_k}(n_k - n_{k-1}) \end{aligned}$$

- **Invariant distribution**

  If the Markov chain is ergodic, then the invariant distribution $\pi = (\pi_0, \pi_1, \pi_2 \ldots, \pi_N)$ is the unique nonnegative solution of

  $$\begin{cases} \pi_y = \sum_{x=0}^{N} \pi_x p_{xy} \quad \text{where} y = 0, 1, 2, \cdots, N, \\ \pi_0 + \pi_1 + \cdots + \pi_N = 1. \end{cases}$$

The end

Thank you !